Project Scope Document

Project Title: AI-Assisted Travel Planning Web Application

1. Project Overview

The proposed project is an AI-assisted web application designed to help users create personalised short-trip travel itineraries.

Users will be able to enter basic travel preferences, such as destination, travel dates, budget, number of travellers and personal interests. The system will use a Large Language Model (LLM) to generate a structured travel itinerary based on these preferences.

The main purpose of the application is to simplify the travel planning process by reducing the amount of time users need to spend searching for, filtering and organising travel information manually.

2. Project Aim

The main aim of the project is to design and develop a usable web application that demonstrates how AI can support users in a practical everyday planning task.

- collecting user travel preferences;
- generating personalised travel itineraries using an LLM;
- presenting generated plans in a clear and user-friendly format;
- exploring methods for reducing unreliable or unsuitable AI-generated outputs;
- evaluating whether the generated plans are useful and relevant to user requirements.

3. Target Users

The intended users are people who want to plan short leisure trips but do not want to spend a large amount of time researching and organising travel information manually.

The initial version of the application will focus on relatively simple short-trip planning rather than complex long-term travel, international transport booking or real-time reservation services.

4. Proposed User Flow

1 The user enters a destination.

2 The user specifies the travel dates.

3 The user provides an approximate budget and number of travellers.

4 The user selects or enters personal interests, such as culture, food, nature, shopping or photography.

5 The system processes the information and generates a personalised itinerary.

6 The itinerary is presented in a structured format, for example by day and by morning, afternoon and evening activities.

7 The system presents supporting information such as a budget summary and basic plan checks or uncertainty indicators where appropriate.

8 The user reviews the generated travel plan.

9 If feasible, the user may edit or save the plan.

5. Initial Functional Requirements

The requirements are prioritised using the MoSCoW method: Must have, Should have, Could have and Won't have for the initial version.

Core Functional Requirements

FR1 - Destination Input (Must have)

The system shall allow users to enter or select a travel destination.

FR2 - Travel Dates Input (Must have)

The system shall allow users to specify the start and end dates of the trip.

FR3 - Budget Input (Must have)

The system shall allow users to provide an approximate travel budget.

FR4 - Interest Preferences (Must have)

The system shall allow users to specify personal interests or preferred activity categories.

FR5 - Itinerary Generation (Must have)

The system shall generate a personalised travel itinerary based on the user's destination, dates, budget, traveller information and interests.

FR6 - Structured Output (Must have)

The system shall present the generated itinerary in a clear and structured format.

FR7 - Preference Matching (Must have)

The generated itinerary should reflect the user preferences provided as input.

FR8 - Error Handling (Should have)

The system shall provide appropriate feedback when required information is missing or when itinerary generation fails.

Possible Additional Requirements

The following features may be considered if the core system is completed within the available project time:

FR9 - Plan Editing (Could have)

Users may be able to edit parts of a generated itinerary.

FR10 - Plan Saving (Could have)

Users may be able to save generated travel plans for later viewing.

FR11 - Curated Attraction Data (Should have)

The system may use a small curated dataset of real attractions to provide more reliable information and reduce the risk of the LLM inventing unsuitable or non-existent locations.

FR12 - Additional Information Integration (Won't have for initial version)

Weather information or simple map-based presentation may be explored as future extensions if technically feasible.

FR13 - Basic Constraint Validation (Should have)

The system should perform basic checks for obvious issues such as whether the estimated plan exceeds the stated budget or contains unrealistic timing conflicts.

FR14 - Uncertainty / Verification Indicators (Should have)

The system should identify information that may change, such as prices or opening hours, and indicate when users should verify it before travelling.

6. Scope Boundaries

To keep the project manageable, the initial system will not aim to provide:

- real-time hotel or flight booking;
- real-time transport ticket purchasing;

- full route optimisation;
- support for every possible destination;
- a complex multi-agent architecture;
- large-scale real-time travel data integration;
- a complete commercial travel booking platform.

These features may be discussed as future work but are not part of the initial project scope.

7. Initial Technical Direction

The project is currently expected to be developed as a responsive web application.

A possible implementation may use a React-based front end, a Python/Flask backend, an LLM API for itinerary generation, structured prompts to control generated outputs, and optional local or lightweight data storage for saved plans or curated attraction information.

The final technology stack may be adjusted during implementation after further discussion with the supervisor.

8. Initial Evaluation Considerations

Because LLM outputs are probabilistic and may vary between requests, evaluation will be an important part of the project.

- checking whether generated itineraries satisfy the input requirements;
- testing whether recommended activities match user interests and budget;
- checking for unsuitable, unrealistic or incorrect travel suggestions;
- checking basic timing and budget consistency where feasible;
- using a fixed set of test cases to compare outputs;
- collecting user feedback on usefulness, clarity and relevance;
- analysing examples of poor or unsuccessful generated plans.

An initial comparison of existing LLM travel-planning outputs has been used to identify issues such as assumptions, variable budget estimates, uncertain dynamic information and inconsistent timing. The final evaluation approach will be refined after discussion with the supervisor.

9. Expected Minimum Viable Product

The initial minimum viable product will aim to include the following core elements.

1 a travel preference input interface;

2 destination, travel dates, budget, traveller and interest inputs;

3 LLM-based itinerary generation;

4 a structured itinerary results page;

5 basic validation, uncertainty indicators and error handling.

Editing, saving plans, curated attraction data, weather information and map presentation will remain extensions unless time allows.